

Ex03. Introduction to Signals and systems

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1.1. Develop 4 discrete signal charts

This assignment involves plotting four discrete-time signals as specified in the problem set. The signals are defined using $\alpha = 0.5$, with the index k ranging from -5 to 5 in increments of 1. To visualize these signals, the stem function was utilized. The specific signals analyzed are the unit step function $x[n] = u[n]$ and the exponential signal $h[n] = \alpha^n u[n]$.

```
alpha = 0.5;
k = -5:5;
x = @(n) (n >= 0);
h = @(n) (alpha.^n) .* (n >= 0);
y = @(n) ((1 - alpha.^(n+1))/(1 - alpha)) .* (n >= 0);

figure('Position', [100, 100, 600, 600]);

subplot(2, 2, 1);
p1 = stem(k, x(k));
title('1. x[k] Signal');
xlabel("Time");
ylabel("Amplitude");
set(p1, 'Color', 'r');
set(p1, 'Marker', '^');
grid on;

subplot(2, 2, 2);
p2 = stem(k, h(-2-k));
title('2. h[-2-k] Signal');
xlabel("Time");
ylabel("Amplitude");
set(p2, 'Color', 'g');
set(p2, 'Marker', '+');
grid on;

subplot(2, 2, 3);
p3 = stem(k, h(3-k));
title('3. h[3-k] Signal');
xlabel("Time");
ylabel("Amplitude");
set(p3, 'Color', 'b');
set(p3, 'Marker', 'x');
grid on;

subplot(2, 2, 4);
p4 = stem(k, y(k));
title('4. y[k] Signal');
```

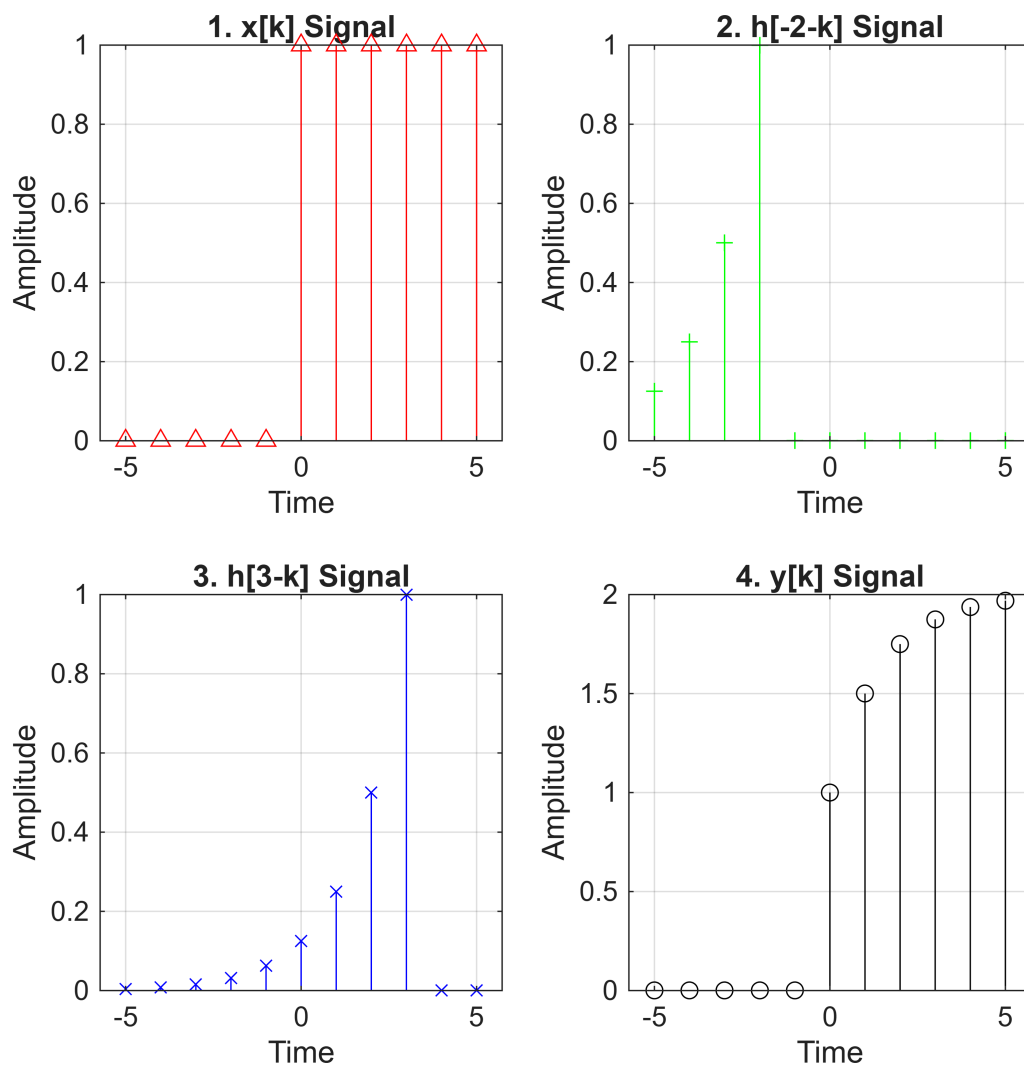
```

xlabel("Time");
ylabel("Amplitude");
set(p4, 'Color', 'k');
set(p4, 'Marker', 'o');
grid on;

```

%% Below is the 4 charts required in the assignemnt.
 %% heaviside function mainly supports to express continuous signal, not
 discrete signal.
 %% (In the case of $x[k]$, it returns 0.5 at time 0 which is not the result of
 an ideal step function)
 %% Then, I used $(n \geq 0)$ as a substitute for that.
 sgtitle('Discrete Signals Analysis');

Discrete Signals Analysis



1.2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

The image shows a photograph of a piece of lined paper with handwritten mathematical expressions. The expressions are as follows:

- Top row: $x[k]h[n-k]$, $1 \times \alpha^{n-k} \times 1$, and α^{n-k} . Each expression is underlined with a single horizontal line and has a diagonal slash mark at the end.
- Bottom row: α^m , α^m , $\frac{1 - \alpha^{n+1}}{1 - \alpha}$. The first two α^m terms are each underlined with a single horizontal line and have a diagonal slash mark. The fraction is underlined with a double horizontal line and has a diagonal slash mark.